

VU TRONG CHAU

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WORK EXPERIENCE

TechX | AI Systems & LLM Engineering Intern

May 2026 - Present

- Engineered the AgentStore AI marketplace using FastAPI, Python, React, and TypeScript, building scalable APIs, tool registries, and web interfaces for AI agent discovery, deployment, and sharing
- Implemented MCP-based tool orchestration, execution tracing, and multi-LLM integrations, improving observability, reliability, and debugging of autonomous workflows across a cross-functional engineering team
- Developed ranking, recommendation, and analytics pipelines using Python, Pandas, and Jupyter, transforming marketplace usage data into actionable insights that improved discoverability and performance evaluation

PROJECTS

Healthcare Chatbot | Python, LangGraph, Langchain, Flask, FAISS, Groq, AWS, Docker, RAGAS | [GitHub](#)

May 2026

- Built a 5-agent LangGraph RAG pipeline over 500K+ medical records, achieving 90% accuracy and 0.898 macro F1-score
- Engineered AWS S3, FAISS vector storage, and ETL pipelines; RAGAS-guided chunking reduced unsafe response rates to <0.3%
- Deployed Dockerized microservices with GitHub Actions CI/CD and Prometheus monitoring for production-ready AI systems

Credit Risk Analytics | Python, Tableau, Scikit-learn, Pandas, NumPy, Flask, FRED API | [GitHub](#)

Dec 2025

- Built end-to-end ML pipelines for loan default prediction using 2.26M+ financial records, processing 110K+ loans across 7 Plains states
- Developed regression and classification models, achieving $R^2=0.91$ and AUC of 0.79 with 89.7% accuracy on imbalanced datasets
- Engineered macroeconomic features and automated ETL pipelines integrating FRED, BLS, and BEA data with zero missing values

Threat Detection using Machine Learning | Python, PyTorch, FastAPI, NLP, Hugging Face, Docker, AWS, CI/CD | [GitHub](#)

Aug 2025

- Built a production-ready AI threat detection platform using 130K+ labeled texts to classify threat content in real time using FastAPI, scikit-learn, and Hugging Face Toxic-BERT, achieving 84.9% accuracy and 0.855 F1-score
- Developed NLP pipelines, REST APIs, and analytics dashboards using TF-IDF, Logistic Regression, NLTK, PyTorch, Pandas, and SQLite
- Deployed the application with Docker, GitHub Actions CI/CD, AWS EC2/S3, and Hugging Face Spaces for automated cloud deployment

Sleep Quality Prediction | Python, Scikit-learn, Flask, Pandas, NumPy | [GitHub](#)

Apr 2025

- Built ML pipelines on 110K+ records to classify sleep disorders and predict sleep quality using multiple machine learning models
- Achieved 82.4% classification accuracy and $R^2=0.67$ on regression tasks by engineering behavioral and physiological health features
- Developed a Flask prediction API and real-time analytics dashboard with personalized health recommendations and deployed ML models

Global Population Prediction | Python, Scikit-learn, Pandas, D3.js, ETL Pipelines, Flask | [GitHub](#)

Dec 2024

- Engineered a ETL pipeline integrating 12 World Bank indicators across 195+ countries and 60+ years (1960–2023) of demographic data
- Developed predictive models achieving $R^2 = 0.945$ while evaluating population growth trends and socioeconomic indicators
- Built an interactive D3.js dashboard featuring choropleth maps, time-series analytics, and country-level demographic comparisons

EDUCATION

Troy University

Troy, Alabama

Master of Science in Computer Science - Artificial Intelligence (GPA: 3.5/4.0)

Jul 2025

Coursework: Analysis of Algorithms, Computer Architecture, Machine Learning, Advanced Artificial Intelligence, Data Visualization

Graduate Coursework in Master of Business Administration (MBA) - Business Analytics

May 2026

University of Sunderland

Sunderland, UK

Bachelor of Engineering in Electronic and Electrical Engineering | UK 2:1 Honours

Jul 2021

TECHNICAL SKILLS

Generative AI & LLMs: RAG, Prompt Engineering, FAISS, Chroma, RAGAS, Fine-tuning, Quantization, MCP, Agentic Pipelines

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face, LangChain, LangGraph, Pandas, NumPy, NLTK

MLOps & Cloud: Docker, Kubernetes, CI/CD, GCP Vertex AI, AWS, FastAPI, Triton Inference Server, Tableau, D3.js, ETL Pipelines

Programming Languages: Python, SQL, C/C++, JavaScript, TypeScript

CERTIFICATIONS

LLM Application Engineering and Development | [Simplilearn](#) | [Link](#)

Oct 2025

Data Science Methodology | [IBM](#) | [Link](#)

Sep 2025

Generative AI with Large Language Models | [DeepLearning.AI](#) | [Link](#)

Sep 2025